

The Future of Hybrid Care and Wellbeing in HCI

Karthik S. Bhat*
Georgia Tech
USA

Azra Ismail*
Georgia Tech
USA

Amanda K. Hall
Microsoft Research
USA

Naveena Karusala
Harvard University
USA

Helena Mentis
University of Maryland, Baltimore
County
USA

John Vines
University of Edinburgh
UK

Neha Kumar
Georgia Tech
USA

ABSTRACT

This workshop focuses on remote care and wellbeing as we transition into a world increasingly adopting hybrid lifestyles and modes of operation. Care and care work have predominantly been researched in traditionally in-person interpersonal contexts. The burgeoning uptake and incorporation of information and communication technologies towards remote care have created new workflows and resulted in emerging questions around the definitions and scope of care practice in response. The confluence of technological, sociocultural, geopolitical, and climatic realities of the current day brings into focus the need to unpack the idea of “care,” and the role that HCI researchers could play in creating equitable futures of remote and hybrid care. This workshop will focus on questions such as “*What does holistic wellbeing look like in the era of hybrid caregiving?*” and “*How does environmental care factor into our research practice?*” We invite researchers and practitioners from academia and industry in this workshop to reflect on these questions and advance the future of remote care work at CHI.

KEYWORDS

hybrid, remote, care, future of work

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1 INTRODUCTION

“On the most general level, we suggest that caring be viewed as a species activity that includes everything

that we do to maintain, continue, and repair our ‘world’ so that we can live in it as well as possible. That world includes our bodies, our selves, and our environment, all of which we seek to interweave in a complex, life-sustaining web.” [12]

The recent years have seen a significant reimagining of what work and workplaces look like, precipitated by the COVID-19 pandemic. In many professions, including healthcare, remote and hybrid work practices have become more prevalent since the onset of the pandemic, with geographic boundaries, commute constraints no longer hindering collaboration and production [1, 6]. These recent changes coexist with, and compound, other longer-term modifications to work practice including the growth of gig work and platformization (e.g., [7, 33]), and approaches towards automation of many forms of vocational work [23, 40]. The COVID-19 pandemic has also changed our workflows and values around work and care responsibilities and our priorities. Not only have we been pushed to question how we live and work, but hybrid modes of operation now allow us to receive care remotely and work remotely. One outcome of this trend is a growing support among HCI researchers for centering workers’ needs and wellbeing in this changing landscape [10, 22, 29, 34], and in the organization of workplaces in the future [38, 39]. With changing technological, economic, geopolitical, and sociocultural realities catalyzed by rapid and slow crises like the COVID-19 pandemic and climate change, there has also been increased public and academic attention towards centering care for the environment in the future of work. As we transition into a post-pandemic world increasingly adopting hybrid lifestyles and modes of operation, a focus on *care* can help us make sense of shifting priorities and how it may be enabled (or not) by the work environment. In light of conversations on building such caring future(s) of work [10, 17], we invite scholars to think about care in an era of remote and hybrid work, by focusing on the activities we do to maintain, continue, and repair our bodies, our selves, and our environment [12].

Much research on care in workplaces in HCI has been in healthcare contexts, with a broad understanding that care work refers to the labor involved in improving the physical and mental health and wellbeing of care recipients (e.g., [41]). Prior research has examined caregiving by formal healthcare workers like nurses [19] and medical teams [13], as well as typically informal care workers

*Both authors contributed equally to this workshop proposal.

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like family members (e.g., [31, 35]) and community health workers [15, 42]. Though these technologies saw slow uptake initially due to legal, financial, and technological barriers, significant strides have been made in recent years, precipitated by the COVID-19 pandemic. Individuals and family caregivers are increasingly seeking support online to gain access to specialized expertise and personalized care. This includes remote care delivery such as telehealth services like chatbots [20, 21], online communities [18, 43], as well as hybrid approaches that combine in-person caregiving with remote support [3]. The growing platformization of healthcare [25, 37] and transition to remote care (e.g., through telemedicine), for example, have created new workflows for care work by redistributing responsibilities among stakeholders in ecologies of care (e.g., [3, 32]). Conversely, new technological and human infrastructures are being created and empowered to support remote care work. Examples include opportunities for care work to turn to gig workers towards supporting remote care [16, 28]; HCI research towards human-AI collaboration in formal and informal care [2, 36]; and the design of technologies to support datafication of frontline health work [15, 30].

Our workshop aims to bring these threads of discussion together—care for our bodies, our selves, and our environment—to chart a path forward for research on the future of hybrid and remote work. Across the threads, the invisibility of care labor—an important consideration in prior research and discourse [8, 9, 42]—would crucially need to be reexamined as sociotechnical care assemblages and infrastructures change with the times. Further, looking beyond care at an individual and community-level, we intend for the workshop to help reflect on what it means to consider the pressing need for environmental and planetary care as goals and outcomes of our research. With climate change being a defining problem of the time, we aim for this discourse to also lead to meaningful insights into a research agenda for sustainable hybrid futures of care work.

2 WORKSHOP THEMES

We intend to welcome relevant themes that our participants propose, in alignment with the goals of our workshop. We will start with inviting conversations on the following themes:

Wellbeing in an Era of Remote Work: This theme will encourage participants to reflect on shifting priorities around work and care, and how the growing transition to remote and hybrid work may affect individual and community wellbeing. Our reflections will include addressing: “How can technology and workplaces for the (hybrid) futures of work be designed to center care?” and “What are the effects of moving work remote on worker wellbeing?”

Remote and Hybrid Caregiving: The goal of this theme is to discuss remote care in the context of personal health and wellbeing, generally exemplified by telemedicine and telehealth [3, 26, 27], patient education and empowerment [5, 14, 24], and health tracking and datafication [4, 11]. We aim to extend conversations in this domain by unpacking how current understandings and approaches to this research could adapt to the increasing adoption of hybrid caregiving practices. We encourage participants to draw connections between their submissions and their worldviews around the hybrid future of care and wellbeing. Our discussions, ideally, will

help answer questions like: “What does holistic wellbeing look like in the era of hybrid caregiving?”

Futures of Environmental Care and Sustainability: Through this theme, we will deliberate on broader dimensions of hybrid care work including how we can draw research attention to environmental care both directly—as a subject of study—and indirectly—as a reflection on the environmental costs of our research. This theme will offer an opportunity to create future research directions on sustainable computing practices in hybrid care work, as well as a meta-level recognition, and plans for amelioration, of the planetary and environmental impacts of our research.

3 WORKSHOP LOGISTICS

3.1 Workshop Participation

We plan to accept 20-25 participants into our workshop. We will advertise calls for participation on mailing lists used by the HCI/CHI community and other relevant disciplines (e.g., Archive of Internet Researchers, Global Digital Health Network), social media spaces like on Twitter and Facebook (e.g., CHI Meta, CSCW Meta), and on relevant Discord Servers and Slack Workspaces from workshops organized in the past around care, wellbeing, and the future of work. The organizers will also individually share this call on their professional and personal social networks, potentially reaching researchers and practitioners outside of HCI. Finally, the call for participation, along with relevant and logistical details, will be uploaded on our workshop website (<https://sites.google.com/view/hybrid-care-wellbeing-chi2023>) following acceptance.

3.2 Pre-Workshop

Our pre-workshop activities will be primarily structured towards creating a shared starting point for discussion. We will create a channel on the SIGCHI Discord server, inviting organizers and accepted participants to begin conversing and sharing relevant resources. These resources could include media like podcasts, videos, and zines, and academic articles relevant to the themes of discussion. All workspace members will be encouraged to share bios and other relevant details about their work to facilitate community building and professional networking. We will use Miro boards to facilitate comments and discussions on all accepted submissions, allowing for asynchronous engagement prior to the workshop. Finally, we will create a Slack channel for participants to note their expectations of the workshop that we will try to incorporate into its structure.

3.3 Workshop

In alignment with the intellectual goals of our workshop, we will support in-person participation as well as remote participation via Zoom. Based on the organizers’ and accepted participants’ modalities of participation, we will create discussion groups—each moderated by at least one organizer—that are either fully in-person or fully remote during the workshop. All discussions would occur synchronously at the time of the workshop, with online participants being video-conferenced into the workshop venue. We will ensure opportunities for discussions between the in-person and remote participants during the workshop. The organizers will review all submissions prior to the workshop and categorize accepted submissions into discussion groups based on the topics of the submissions.

We will assign participants in-person or online groups at the start of the workshop.

Opening and Introductions (1 hour): We will begin by introducing the organizers of the workshop, and our topics of discussion to all the attendees. Following this, we will do a quick round of introductions within discussion groups for participants to get to know each other. In the final 20 minutes, we will have 3-4 participants—chosen based on their submissions—present on their research, to set the stage for following discussions.

Small Group Discussions (2 hours): We will provide all groups with discussion prompts that draw on the themes and questions described earlier, as well as emerging themes from the accepted submissions. They will also be asked to develop plans of action for future work in this space at a personal and collective level. At least one organizer will be part of each discussion group, and will serve as the facilitator and note-taker to maintain a record of discussions within the groups. After 50 minutes, we will reconvene a discussion with the whole group from 10 minutes where we will summarize discussions within the smaller groups. We will repeat this process for the next hour.

Big Group Discussion and Closing Remarks (1 hour): The groups will reconvene into a larger group discussion where we will encourage participants share outcomes of the discussions, including identifying potential future research directions. We will encourage participants from other groups to ask questions to spark further conversations beyond the workshop.

3.4 Post-Workshop Plans

We will summarize and share the notes from the group discussions with the participants after the workshop. Attendees will also be invited to contribute to a Medium blog post and an article in ACM Interactions reflecting on the group discussions. We will share the blog post with the HCI community over social media (Twitter) and on our website. To keep the community engaged beyond the workshop, we will also create a Slack or Discord space for the workshop attendees, which will also be open to their colleagues or collaborators interested in care work to enable further discussion.

4 ORGANIZERS

All organizers have been involved in research on care and the future of work for several years and across multiple geographical contexts. In organizing this workshop, we bring our diverse domain expertise and methodological backgrounds towards facilitating meaningful discussions. More details about the organizers' backgrounds are available below:

Karthik S. Bhat is a PhD candidate in Human-Centered Computing at Georgia Tech. His research takes an ecological perspective to caregiving and investigates how emerging technologies like conversational agents, telemedicine, and wearable sensors can be leveraged towards supporting and augmenting caregiving relationships.

Azra Ismail is a PhD candidate in Human-Centered Computing at Georgia Tech. Her research focuses on the design of data-driven technologies for maternal and child care delivery by workers in India, amidst rapid digitization and datafication of work, and changing gendered norms around digital access.

Amanda Hall is a Senior User Researcher at Microsoft Research in Health Futures on the Medical Experiences and Design Team and an affiliate assistant professor at the University of Washington, Department of Biomedical Informatics and Medical Education. Her research focuses on the design and development of technology (hardware and software) for improved medical decision making and health behavior change. She has over 6 years of experience working in home healthcare doing patient care coordination and leading telehealth programs.

Naveena Karusala is a Postdoctoral Fellow at Harvard's Center for Research on Computation and Society. Her research is at the intersection of care, labor, and emerging technologies, spanning domains such as health, safety, and social work.

Helena Mentis is a professor in the Department of Information Systems at the University of Maryland, Baltimore County (UMBC). She investigates situated communication and collaboration practices in healthcare, and, in turn, develops and deploys new mechanisms for collaboratively sensing, presenting, and interacting with information.

John Vines is a Professor in the School of Informatics, University of Edinburgh, and co-directs the Institute for Design Informatics. His research focuses on participatory and co-creative approaches to technology design, and he has conducted projects at the intersection of ageing, care and wellbeing for over 15 years.

Neha Kumar is an associate professor at Georgia Tech, conducting research at the intersection of human-centered computing and global development. Much of her work focuses on global health and the role that technology might play in enabling equitable, healthier futures of frontline health work.

5 CALL FOR PARTICIPATION

The goal of this workshop is to discuss remote care and wellbeing as we transition into a world increasingly adopting hybrid lifestyles and modes of operation. This workshop will be a venue to discuss questions such as “*What does holistic wellbeing look like in the era of hybrid caregiving?*”, “*How does environmental care factor into our research practice?*” among others around hybrid care and wellbeing. We invite researchers and practitioners from academia and industry to reflect on these questions and advance the future of remote and hybrid care at CHI.

We invite submissions in the form of short papers (2-4 pages excluding references) or abstracts (300 words). All submissions will be due on Feb 23, 2023, and need to be formatted according to the ACM Master template (single column). Submissions should ideally be focused on remote and hybrid care, future of care work, and related topics. The workshop themes can serve as a guide, and we encourage submissions that go beyond the ideas covered within them on these topics. Submissions may present current or past research projects, or could be position papers, literature reviews, or theoretical analyses.

All submissions will be reviewed by the organizers, and accepted submissions will be posted on our website (<https://sites.google.com/view/hybrid-care-wellbeing-chi2023>). Please note that at least one author of each accepted submission must attend the workshop—we

will support both remote and in-person participation—and all participants be registered for both the workshop and at least one day of the conference. Please direct any questions to ksbhat@gatech.edu.

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REFERENCES

- [1] Cevat Giray Aksoy, Jose Maria Barrero, Nicholas Bloom, Steven J Davis, Mathias Dolls, and Pablo Zarate. 2022. *Working from home around the world*. Technical Report. National Bureau of Economic Research.
- [2] Vince Bartle, Janice Lyu, Freesoul El Shabazz-Thompson, Yunmin Oh, Angela Anqi Chen, Yu-Jan Chang, Kenneth Holstein, and Nicola Dell. 2022. "A Second Voice": Investigating Opportunities and Challenges for Interactive Voice Assistants to Support Home Health Aides. In *CHI Conference on Human Factors in Computing Systems*. 1–17.
- [3] Karthik S. Bhat, Mohit Jain, and Neha Kumar. 2021. Infrastructuring Telehealth in (In)Formal Patient-Doctor Contexts. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 323 (Oct. 2021), 28 pages. <https://doi.org/10.1145/3476064>
- [4] Karthik S. Bhat and Neha Kumar. 2020. Sociocultural Dimensions of Tracking Health and Taking Care. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 129 (Oct. 2020), 24 pages. <https://doi.org/10.1145/3415200>
- [5] Timothy W. Bickmore, Laura M. Pfeifer, and Brian W. Jack. 2009. Taking the Time to Care: Empowering Low Health Literacy Hospital Patients with Virtual Nurse Agents. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). ACM, New York, NY, USA, 1265–1274. <https://doi.org/10.1145/1518701.1518891>
- [6] Erik Brynjolfsson, John J Horton, Adam Ozimek, Daniel Rock, Garima Sharma, and Hong-Yi TuYe. 2020. *COVID-19 and remote work: An early look at US data*. Technical Report. National Bureau of Economic Research.
- [7] Martha Alter Chen. 2008. *Informalization of Labour Markets: Is Formalization the Answer?* Routledge.
- [8] Yunan Chen, Victor Ngo, and Sun Young Park. 2013. Caring for Caregivers: Designing for Integrality. In *Proceedings of the 2013 Conference on Computer Supported Cooperative Work* (San Antonio, Texas, USA) (CSCW '13). Association for Computing Machinery, New York, NY, USA, 91–102. <https://doi.org/10.1145/2441776.2441789>
- [9] Yunan Chen, Nervo Verdezoto, Xinning Gui, Xiaojuan Ma, Claus Bossen, Naveen Bagalkot, Valeria Herskovic, and Bernd Ploderer. 2019. Unpacking the Infrastructuring Work of Patients and Caregivers around the World. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3290607.3299021>
- [10] CHIWORK Collective, Naveena Karusala, Nabil Al Nahin Ch, Diana Tosca, Alberta A. Ansah, Emeline Brulé, Nadia Fereydooni, Le-En Huang, Azra Ismail, Pranjal Jain, Yi Xuan Khoo, Isabel Muñoz, Clemens Schartmüller, Himanshu Verma, Preeti Vyas, Susanne CJ Boll, Sarah E Fox, Noopur Raval, Max L Wilson, Anna L Cox, Christian P Janssen, Helena M Mentis, Neha Kumar, Orit Shaer, and Andrew L Kun. 2022. Human-Computer Interaction and the Future of Work. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 162, 3 pages. <https://doi.org/10.1145/3491101.3516407>
- [11] Mayara Costa Figueiredo. 2021. *Data Work and Data Tracking Technologies in Fertility Care: A Holistic Approach*. University of California, Irvine.
- [12] Berenice Fisher and Joan Tronto. 1990. Toward a feminist theory of caring. *Circles of care: Work and identity in women's lives* (1990), 35–62.
- [13] Geraldine Fitzpatrick and Gunnar Ellingsen. 2013. A Review of 25 Years of CSCW Research in Healthcare: Contributions, Challenges and Future Agendas. *Comput. Supported Coop. Work* 22, 4–6 (aug 2013), 609–665. <https://doi.org/10.1007/s10606-012-9168-0>
- [14] Matthew K. Hong, Clayton Feustel, Meeshu Agnihotri, Max Silverman, Stephen F. Simoneaux, and Lauren Wilcox. 2017. Supporting Families in Reviewing and Communicating About Radiology Imaging Studies. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). ACM, New York, NY, USA, 5245–5256. <https://doi.org/10.1145/3025453.3025754>
- [15] Azra Ismail, Naveena Karusala, and Neha Kumar. 2018. Bridging Disconnected Knowledges for Community Health. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW, Article 75 (Nov. 2018), 27 pages. <https://doi.org/10.1145/3274344>
- [16] Shane Jackson. 2022. Covid, Healthcare Workers And The Rise Of The Gig Economy—Lessons For Every Industry. <https://www.forbes.com/sites/forbesbusinesscouncil/2022/07/28/covid-healthcare-workers-and-the-rise-of-the-gig-economy-lessons-for-every-industry/>
- [17] Naveena Karusala, Azra Ismail, Karthik S Bhat, Aakash Gautam, Sachin R Pendse, Neha Kumar, Richard Anderson, Madeline Balaam, Shaowen Bardzell, Nicola J Bidwell, Melissa Densmore, Elizabeth Kazianas, Anne Marie Piper, Noopur Raval, Pushpendra Singh, Austin Toombs, Nervo Verdezoto, and Ding Wang. 2021. The Future of Care Work: Towards a Radical Politics of Care in CSCW Research and Practice. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing* (Virtual Event, USA) (CSCW '21). Association for Computing Machinery, New York, NY, USA, 338–342. <https://doi.org/10.1145/3462204.3481734>
- [18] Naveena Karusala, David Odhiambo Seeh, Cyrus Mugo, Brandon Guthrie, Megan A Moreno, Grace John-Stewart, Irene Inwani, Richard Anderson, and Keshet Ronen. 2021. "That Courage to Encourage": Participation and Aspirations in Chat-Based Peer Support for Youth Living with HIV. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 223, 17 pages. <https://doi.org/10.1145/3411764.3445313>
- [19] Naveena Karusala, Ding Wang, and Jacki O'Neill. 2020. Making Chat at Home in the Hospital: Exploring Chat Use by Nurses. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3313831.3376166>
- [20] Ahmet Baki Kocaballi, Shlomo Berkovsky, Juan C. Quiroz, Liliana Laranjo, Huong Ly Tong, Dana Rezazadegan, Agustina Briatore, and Enrico Coiera. 2019. The Personalization of Conversational Agents in Health Care: Systematic Review. *Journal of Medical Internet Research* 21, 11 (2019), e15360. <https://doi.org/10.2196/15360>
- [21] A. Baki Kocaballi, Juan C. Quiroz, Liliana Laranjo, Dana Rezazadegan, Rafal Kocielnik, Leigh Clark, Q. Vera Liao, Sun Young Park, Robert J. Moore, and Adam Miner. 2020. Conversational Agents for Health and Wellbeing. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI EA '20). Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3334480.3375154>
- [22] Siân Lindley, Noopur Raval, Hamed S Alavi, Silvia Lindtner, and Ding Wang. 2019. The future of work. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [23] Susan Lund, Anu Madgavkar, James Manyika, Sven Smit, Kweilin Ellingrud, Mary Meaney, and Olivia Robinson. 2021. *The future of work after COVID-19*. Technical Report. McKinsey Global Institute.
- [24] Haley MacLeod, Grace Bastin, Leslie S. Liu, Katie Siek, and Kay Connelly. 2017. "Be Grateful You Don't Have a Real Disease": Understanding Rare Disease Relationships. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). ACM, New York, NY, USA, 1660–1673. <https://doi.org/10.1145/3025453.3025796>
- [25] Lital Marom. 2021. Healthcare Revolution In The Platform Economy. <https://www.forbes.com/sites/forbescoachescouncil/2021/03/01/healthcare-revolution-in-the-platform-economy/>
- [26] Helena M. Mentis, Amine Chellali, and Steven Schwaartzberg. 2014. Learning to See the Body: Supporting Instructional Practices in Laparoscopic Surgical Procedures. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 2113–2122. <https://doi.org/10.1145/2556288.2557387>
- [27] Helena M. Mentis, Ahmed Rahim, and Pierre Theodore. 2016. Crafting the Image in Surgical Telemedicine. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing* (San Francisco, California, USA) (CSCW '16). Association for Computing Machinery, New York, NY, USA, 744–755. <https://doi.org/10.1145/2818048.2819978>
- [28] Geoffrey Nagle. 2021. Understanding Gig Workers: Disruptors Of Healthcare Or An Untapped Market. <https://www.forbes.com/sites/forbesbusinessdevelopmentcouncil/2021/08/20/understanding-gig-workers-disruptors-of-healthcare-or-an-untapped-market/>
- [29] Joseph W Newbold, Anna Rudnicka, David Cook, Marta E Cecchinato, Sandy JJ Gould, and Anna L Cox. 2022. The new normals of work: a framework for understanding responses to disruptions created by new futures of work. *Human-Computer Interaction* (2022), 1–24.
- [30] Chinasa T. Okolo, Srjana Kamath, Nicola Dell, and Aditya Vashistha. 2021. "It Cannot Do All of My Work": Community Health Worker Perceptions of AI-Enabled Mobile Health Applications in Rural India. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 701, 20 pages. <https://doi.org/10.1145/3411764.3445420>
- [31] Laura R. Pina, Sang-Wha Sien, Teresa Ward, Jason C. Yip, Sean A. Munson, James Fogarty, and Julie A. Kientz. 2017. From Personal Informatics to Family Informatics: Understanding Family Practices Around Health Monitoring. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* (Portland, Oregon, USA) (CSCW '17). ACM, New York, NY, USA, 2300–2315. <https://doi.org/10.1145/2998181.2998362>
- [32] Divya Raj and TK Srikanth. 2021. Assisted Telemedicine Model for Rural Healthcare Ecosystem. In *13th ACM Web Science Conference 2021*. ACM, Virtual Event

- United Kingdom, 86–91. <https://doi.org/10.1145/3462741.3466649>
- [33] Noopur Raval and Paul Dourish. 2016. Standing out from the crowd: Emotional labor, body labor, and temporal labor in ridesharing. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*. 97–107.
- [34] Anna Rudnicka, Dave Cook, Marta E Cecchinato, Sandy JJ Gould, Joseph W Newbold, and Anna L Cox. 2022. The end of the active work break? Remote work, sedentariness and the role of technology in creating active break-taking norms. In *2022 Symposium on Human-Computer Interaction for Work*. 1–13.
- [35] Maren Schorch, Lin Wan, David William Randall, and Volker Wulf. 2016. Designing for Those Who Are Overlooked: Insider Perspectives on Care Practices and Cooperative Work of Elderly Informal Caregivers. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing* (San Francisco, California, USA) (CSCW '16). ACM, New York, NY, USA, 787–799. <https://doi.org/10.1145/2818048.2819999>
- [36] Mark Schurgin, Mark Schlager, Laura Vardoulakis, Laura R. Pina, and Lauren Wilcox. 2021. *Isolation in Coordination: Challenges of Caregivers in the USA*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445413>
- [37] Michael Serbinis. 2022. The Platformization Of Healthcare Is Here. <https://www.forbes.com/sites/forbestechcouncil/2022/06/08/the-platformization-of-healthcare-is-here/>
- [38] Jaime Teevan, Nancy Baym, Jenna Butler, Brent Hecht, Sonia Jaffe, Kate Nowak, Abigail Sellen, Longqi Yang, Marcus Ash, Kagonya Awori, Mia Bruch, Piali Choudhury, Adam Coleman, Scott Counts, Shiraz Cupala, Mary Czerwinski, Ed Doran, Elizabeth Fetterolf, Mar Gonzalez Franco, Kunal Gupta, Aaron L Halfaker, Constance Hadley, Brian Houck, Kori Inkpen, Shamsi Iqbal, Eric Knudsen, Stacey Levine, Siân Lindley, Jennifer Neville, Jacki O'Neill, Rick Pollak, Victor Poznanski, Sean Rintel, Neha Parikh Shah, Siddharth Suri, Adam D. Troy, and Mengting Wan. 2022. *Microsoft New Future of Work Report 2022*. Technical Report MSR-TR-2022-3. Microsoft. <https://www.microsoft.com/en-us/research/publication/microsoft-new-future-of-work-report-2022/>
- [39] Thomaz Teodorovicz, Raffaella Sadun, Andrew L Kun, and Orit Shaer. 2021. How does working from home during COVID-19 affect what managers do? Evidence from time-Use studies. *Human-Computer Interaction* (2021), 1–26.
- [40] Divy Thakkar, Neha Kumar, and Nithya Sambasivan. 2020. Towards an AI-powered future that works for vocational workers. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [41] Austin Toombs, Laura Devendorf, Patrick Shih, Elizabeth Kaziunas, David Nemer, Helena Mentis, and Laura Forlano. 2018. Sociotechnical Systems of Care. In *Companion of the 2018 ACM Conference on Computer Supported Cooperative Work and Social Computing* (Jersey City, NJ, USA) (CSCW '18). Association for Computing Machinery, New York, NY, USA, 479–485. <https://doi.org/10.1145/3272973.3273010>
- [42] Nervo Verdezoto, Naveen Bagalkot, Syeda Zainab Akbar, Swati Sharma, Nicola Mackintosh, Deirdre Harrington, and Paula Griffiths. 2021. The Invisible Work of Maintenance in Community Health: Challenges and Opportunities for Digital Health to Support Frontline Health Workers in Karnataka, South India. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 91 (apr 2021), 31 pages. <https://doi.org/10.1145/3449165>
- [43] Deepika Yadav, Kirti Dabas, Prerna Malik, Anushka Bhandari, and Pushpendra Singh. 2022. “Should I Visit the Clinic”: Analyzing WhatsApp-Mediated Online Health Support for Expectant and New Mothers in Rural India. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 296, 20 pages. <https://doi.org/10.1145/3491102.3517575>